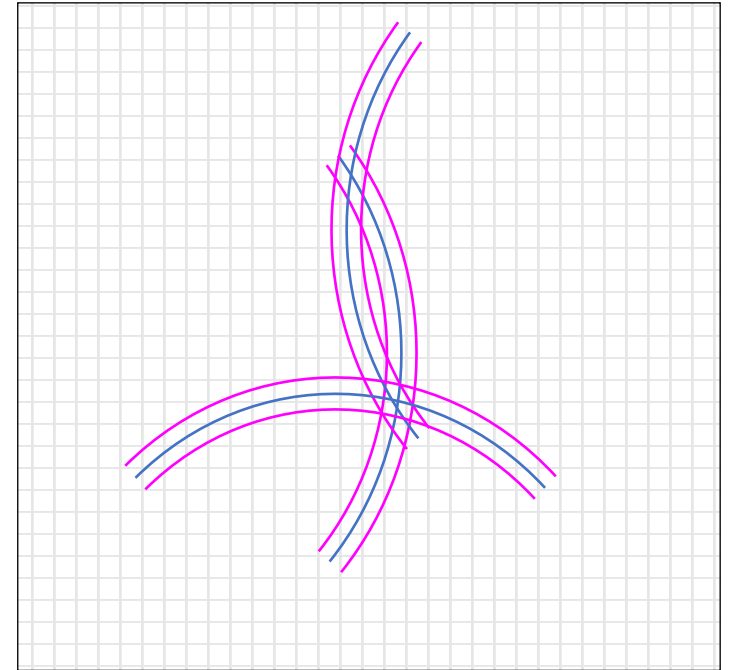
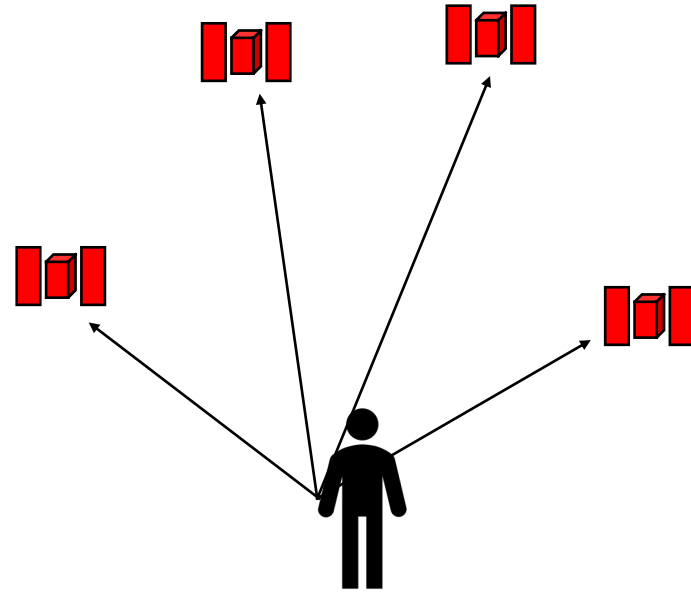


ASEN 5090 Lecture 13

GPS Least Squares Point Solutions



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Course schedule

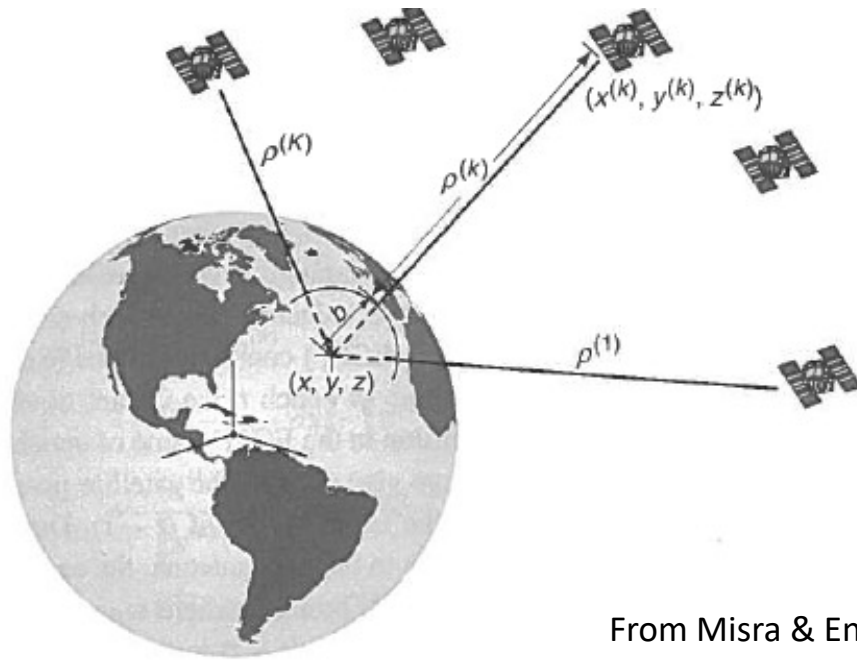
Week	Date	Topic	Reading	Assignments - Due 11pm*
7	10/5/26	L12. Least Squares Solutions	ME Ch 6.A	
	10/7/26	L13. GPS Point Solutions	ME Ch 6.1	
	10/9/26			
8	10/12/26	L14. Improved GPS Solutions		
	10/14/26	L15. Midterm Review & HW Discussion		
	10/16/26			HW5 Due, HW6 Assigned
9	10/19/26	L16. Precise Positioning Methods	ME Ch 7	
	10/21/26	Midterm Exam (no lecture)		Midterm Release 10/21 8AM
	10/23/26			Midterm Close 10/25 11PM
10	10/26/26	L17. Introduction to Signals & Spectra	ME Ch 8	
	10/28/26	L18. GNSS Spectrum	ME Ch 9	
	10/30/26			HW6 Due, HW7 Assigned

GPS Solution with Least Squares

“Raw” Range Measurement Equations:

$$\rho^{(k)}(t) = r^{(k)}(t, t - \tau) + c(\delta t_u(t) - \delta t^{(k)}(t - \tau)) + I^{(k)}(t) + T^{(k)}(t) + \varepsilon^{(k)}(t)$$

$k=1, 2, \dots, K$



From Misra & Enge

1. Satellite clock offset relative to GPST
2. Relativity correction
3. Ionospheric delay using Klobuchar model or eliminate with dual frequency
4. Tropospheric delay using model
5. Carrier smoothing of pseudorange to reduce multipath/noise effects
6. SV orbit/clock correction using local- or wide-area DGPS services
7. ...

GPS Measurement Model

$$\rho_1 = R + c(\delta t_u - \delta t^s) + T + I_{\rho 1} + M_{\rho 1} + \varepsilon_{\rho 1}$$

$$\rho_2 = R + c(\delta t_u - \delta t^s) + T + I_{\rho 2} + M_{\rho 2} + \varepsilon_{\rho 2}$$

ρ_1 = pseudorange measured on L1 frequency based on code

ρ_2 = pseudorange measured on L2 frequency based on code

R = geometrical range from satellite s to user u

δt_u = user/receiver clock error

δt^s = satellite clock error

T = tropospheric delay

$I_{\rho 1/2}$ = ionospheric delay in code measurement on L1/2

$M_{\rho 1/2}$ = multipath delay in code measurement on L1/2

$\varepsilon_{\rho 1/2}$ = other delay/errors in code measurement on L1/2

GPS Measurement Model

$$\rho = R + b + \varepsilon_0 + \varepsilon_\rho$$

ε_0 = are the known or modeled delay/errors in code measurement

$$R = \sqrt{(x^s - x)^2 + (y^s - y)^2 + (z^s - z)^2}$$

GPS measurement is nonlinearly related to the state we want to solve for:



Recall the linear model...

$$y = Ax + \varepsilon$$

where

y is an $[m \times 1]$ column matrix of the measurements

x is an $[n \times 1]$ column matrix of the state elements to be estimated

A is an $[m \times n]$ measurement sensitivity matrix

ε is an $[m \times 1]$ column matrix of the misfits or error,

each assumed to be zero mean with variance σ^2



Linearize the measurement model – Taylor Expansion

$$f(\mathbf{x}) = f(\mathbf{x}_0) + \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}=\mathbf{x}_0} (\mathbf{x} - \mathbf{x}_0) + H.O.T. \approx f(\mathbf{x}_0) + \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta \mathbf{x}$$

$$\rho(\mathbf{x}) = \sqrt{(x^{(s)} - x)^2 + (y^{(s)} - y)^2 + (z^{(s)} - z)^2} + b + \varepsilon_0 + \varepsilon_\rho$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \\ b \end{bmatrix}$$

$$\rho(\mathbf{x}) \approx \rho(\mathbf{x}_0) + \left. \frac{\partial \rho}{\partial x} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta x + \left. \frac{\partial \rho}{\partial y} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta y + \left. \frac{\partial \rho}{\partial z} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta z + \left. \frac{\partial \rho}{\partial b} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta b$$

Linearization of pseudorange equation

$$\rho(\mathbf{x}) \approx \rho(\mathbf{x}_0) + \left. \frac{\partial \rho}{\partial x} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta x + \left. \frac{\partial \rho}{\partial y} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta y + \left. \frac{\partial \rho}{\partial z} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta z + \left. \frac{\partial \rho}{\partial b} \right|_{\mathbf{x}=\mathbf{x}_0} \Delta b$$

$$\rho_0 = \sqrt{(x^s - x_0)^2 + (y^s - y_0)^2 + (z^s - z_0)^2} + b_0 + \varepsilon_0$$

This is predicted pseudorange

$$\left. \frac{\partial \rho}{\partial x} \right|_{\mathbf{x}=\mathbf{x}_0} =$$

$$\left. \frac{\partial \rho}{\partial y} \right|_{\mathbf{x}=\mathbf{x}_0} =$$

$$\left. \frac{\partial \rho}{\partial z} \right|_{\mathbf{x}=\mathbf{x}_0} =$$

$$\left. \frac{\partial \rho}{\partial b} \right|_{\mathbf{x}=\mathbf{x}_0} =$$



Linearization of pseudorange equation

$$\delta\rho^{(s)} = \rho^{(s)} - \rho_0^{(s)} = \left[-\frac{x^{(s)} - x_0}{R_0^{(s)}} \quad -\frac{y^{(s)} - y_0}{R_0^{(s)}} \quad -\frac{z^{(s)} - z_0}{R_0^{(s)}} \quad 1 \right] \begin{bmatrix} \delta x \\ \delta y \\ \delta z \\ \delta b \end{bmatrix} + \varepsilon^{(s)}$$

↑
Solve for corrections
to a priori

Notice that G depends on assumed
[x_0 y_0 z_0]

Geometry Matrix

$$G = \begin{bmatrix} -\frac{x^{(1)} - x_0}{R_0^{(1)}} & -\frac{y^{(1)} - y_0}{R_0^{(1)}} & -\frac{z^{(1)} - z_0}{R_0^{(1)}} & 1 \\ -\frac{x^{(2)} - x_0}{R_0^{(2)}} & -\frac{y^{(2)} - y_0}{R_0^{(2)}} & -\frac{z^{(2)} - z_0}{R_0^{(2)}} & 1 \\ -\frac{x^{(3)} - x_0}{R_0^{(3)}} & -\frac{y^{(3)} - y_0}{R_0^{(3)}} & -\frac{z^{(3)} - z_0}{R_0^{(3)}} & 1 \\ \vdots & \vdots & \vdots & \vdots \\ -\frac{x^{(m)} - x_0}{R_0^{(m)}} & -\frac{y^{(m)} - y_0}{R_0^{(m)}} & -\frac{z^{(m)} - z_0}{R_0^{(m)}} & 1 \end{bmatrix}$$

$$\delta\boldsymbol{\rho} = G \begin{bmatrix} \delta\mathbf{r} \\ \delta b \end{bmatrix}$$



Pseudorange Point Positioning

- Least squares estimate

$$\min \left\| \delta \mathbf{p} - \mathbf{G} \begin{bmatrix} \delta \hat{\mathbf{x}} \\ \delta \hat{b} \end{bmatrix} \right\|$$

$$\begin{bmatrix} \delta \hat{\mathbf{x}} \\ \delta \hat{b} \end{bmatrix} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \delta \mathbf{p}$$

- Update $\begin{matrix} \hat{\mathbf{x}} = \mathbf{x}_0 + \delta \hat{\mathbf{x}} \\ \hat{b} = b_0 + \delta \hat{b} \end{matrix} \left. \vphantom{\begin{matrix} \hat{\mathbf{x}} = \mathbf{x}_0 + \delta \hat{\mathbf{x}} \\ \hat{b} = b_0 + \delta \hat{b} \end{matrix}} \right\} \begin{matrix} \text{Use these as new estimates, repeat the process} \\ \text{until the changes in estimates are small enough} \end{matrix}$
- Iterate until converged to get “Point Solution”
- If a priori (initial) estimate is close to correct, iteration is not required



GPS LEAST SQUARES POSITIONING

1. Get at least 4 pseudorange (code) measurements from the receiver
2. Guess your location and receiver clock bias

For each observation at a given time (epoch)

3. Compute the expected range for each satellite using information from broadcast ephemeris and your assumed location
4. Compute pseudorange corrections - satellite clock error, relativity, iono, tropo
5. Compute pre-fit measurement residual (scalar)
6. Compute measurement sensitivity matrix (1×4)

Use all the observations to solve for position correction

7. Assemble measurement residuals sensitivities into arrays
8. Solve for a correction to assumed location and clock bias

If necessary, repeat steps 3-8!

The coordinate frame you choose in defining the G matrix, determines the coordinate frame of your solution.



QUESTIONS

- What accuracy is required for the initial or a priori estimate?
- How does it impact the least squares solution?
- When you are done iterating, what does $\delta\rho$ look like?



Point Solution Errors

Define $\Delta \mathbf{x} = \hat{\mathbf{x}} - \mathbf{x}$, $\Delta b = \hat{b} - b$

$$\begin{bmatrix} \Delta \mathbf{x} \\ \Delta b \end{bmatrix} = (G^T G)^{-1} G^T \Delta \boldsymbol{\rho} = (G^T G)^{-1} G^T \boldsymbol{\varepsilon}_\rho$$

where $E[\boldsymbol{\varepsilon}_\rho] = 0$ $\text{Cov}[\boldsymbol{\varepsilon}_\rho] = E[\boldsymbol{\varepsilon}_\rho \boldsymbol{\varepsilon}_\rho^T] = R$

Then $E \begin{bmatrix} \Delta \mathbf{x} \\ \Delta b \end{bmatrix} = 0$

$$P = E \begin{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta b \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x}^T & \Delta b \end{bmatrix} \end{bmatrix} = (G^T G)^{-1} G^T R G (G^T G)^{-1}$$

$$\text{If } R = \sigma^2 I \quad P = \sigma^2 (G^T G)^{-1}$$

RMS Position Error

$$\text{Let } \mathbf{H} = (\mathbf{G}^T \mathbf{G})^{-1} = \begin{bmatrix} H_{11} & \bullet & \bullet & \bullet \\ \bullet & H_{22} & \bullet & \bullet \\ \bullet & \bullet & H_{33} & \bullet \\ \bullet & \bullet & \bullet & H_{44} \end{bmatrix}$$

$$\text{Cov} \left\{ \begin{bmatrix} \Delta \hat{\mathbf{x}} \\ \Delta \hat{b} \end{bmatrix} \right\} = \sigma^2 (\mathbf{G}^T \mathbf{G})^{-1} = \sigma^2 \mathbf{H} = \begin{bmatrix} \sigma^2 H_{11} & \bullet & \bullet & \bullet \\ \bullet & \sigma^2 H_{22} & \bullet & \bullet \\ \bullet & \bullet & \sigma^2 H_{33} & \bullet \\ \bullet & \bullet & \bullet & \sigma^2 H_{44} \end{bmatrix}$$

$$\sigma_x^2 = \sigma^2 H_{11} \quad \sigma_y^2 = \sigma^2 H_{22} \quad \sigma_z^2 = \sigma^2 H_{33} \quad \sigma_b^2 = \sigma^2 H_{44}$$

$$\sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2} = \sigma \sqrt{H_{11} + H_{22} + H_{33}}$$



Dilution of Precision (DOP)

Position dilution of precision

$$PDOP = \sqrt{H_{11} + H_{22} + H_{33}}$$

Time dilution of precision

$$TDOP = \sqrt{H_{44}}$$

Geometric dilution of precision

$$GDOP = \sqrt{H_{11} + H_{22} + H_{33} + H_{44}}$$

3-D position RMS

$$\sigma \cdot PDOP$$

15

Clock bias RMS

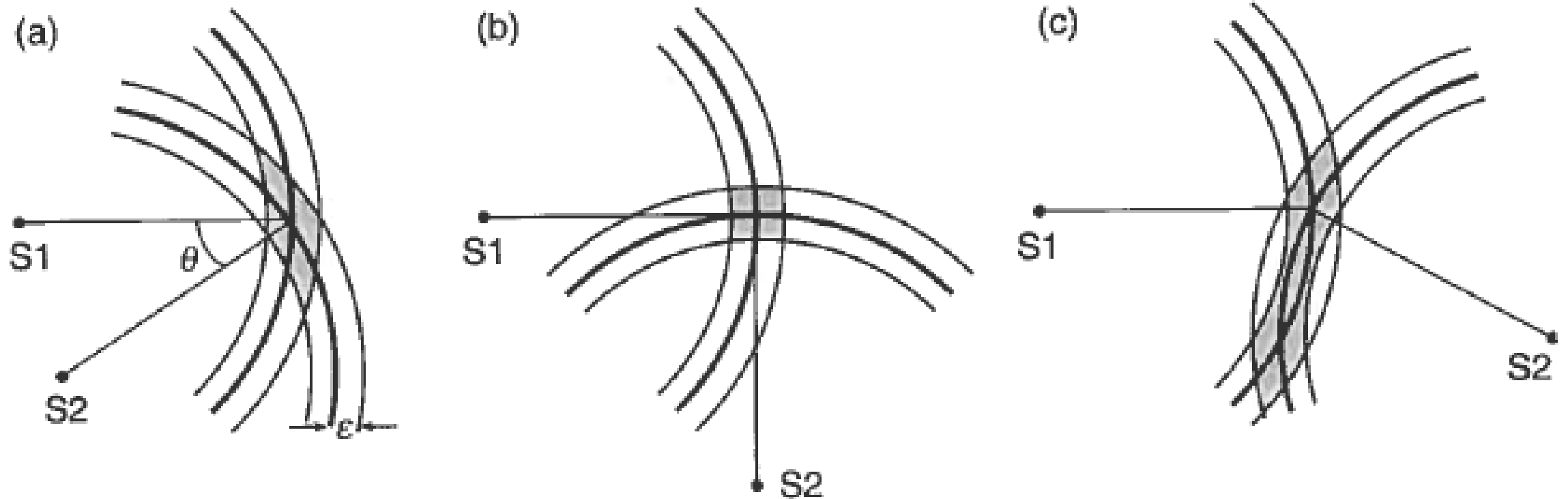
$$\sigma \cdot TDOP$$

3-D position and clock bias RMS

$$\sigma \cdot GDOP$$



Satellite Geometry on Position Error



Position Error in ENU Coordinates

Conversion from ECEF to ENU: $[\Delta \mathbf{x}]_{ENU} = C_{ENU}^{WGS} [\Delta \mathbf{x}]_{WGS}$

$$\Delta \mathbf{x}_L = \mathbf{R}_L \Delta \mathbf{x} \quad \mathbf{R}_L = \begin{bmatrix} -\sin \lambda & \cos \lambda & 0 \\ -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi \end{bmatrix}$$

$$\begin{bmatrix} \Delta \mathbf{x}_L \\ \Delta b \end{bmatrix} = \begin{bmatrix} \mathbf{R}_L & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{x} \\ \Delta b \end{bmatrix} = \tilde{\mathbf{R}}_L \begin{bmatrix} \Delta \mathbf{x} \\ \Delta b \end{bmatrix}$$



Rotating the Covariance Matrix

- Solution will be in WGS-84 coordinates
- It is more useful to consider errors in Local Level Coordinates

$$C_{ENU}^{WGS} = C1(90 - \varphi)C3(90 + \lambda)$$

$$C_{ENU}^{WGS} = \begin{bmatrix} -\sin \lambda & \cos \lambda & 0 \\ -\cos \lambda \sin \varphi & -\sin \lambda \sin \varphi & \cos \varphi \\ \cos \lambda \cos \varphi & \sin \lambda \cos \varphi & \sin \varphi \end{bmatrix}$$

$$P_{ENU} = C_{ENU}^{WGS} P_{WGS} (C_{ENU}^{WGS})^T$$



DOP in ENU Coordinates

$$\tilde{\mathbf{H}} = (\tilde{\mathbf{G}}^T \tilde{\mathbf{G}})^{-1} = \begin{bmatrix} \tilde{H}_{11} & \bullet & \bullet & \bullet \\ \bullet & \tilde{H}_{22} & \bullet & \bullet \\ \bullet & \bullet & \tilde{H}_{33} & \bullet \\ \bullet & \bullet & \bullet & \tilde{H}_{44} \end{bmatrix}$$

Horizontal dilution of precision $HDOP = \sqrt{\tilde{H}_{11} + \tilde{H}_{22}}$

Vertical dilution of precision $VDOP = \sqrt{\tilde{H}_{33}}$

Horizontal position error RMS $\sigma \cdot HDOP$

Vertical error RMS $\sigma \cdot VDOP$

3-D position error RMS $\sigma \cdot PDOP$



DOPS

$$P_{ENU} = C_{ENU}^{WGS} P_{WGS} (C_{ENU}^{WGS})^T$$

$$P_{ENU} = \sigma^2 H_{ENU}$$

$$H_{ENU} = \begin{bmatrix} EDOP^2 & & & \\ & NDOP^2 & & \\ & & VDOP^2 & \\ & & & TDOP^2 \end{bmatrix}$$

$$HDOP = \sqrt{H_{ENU}(1,1) + H_{ENU}(2,2)}$$

$$PDOP = \sqrt{H(1,1) + H(2,2) + H(3,3)}$$

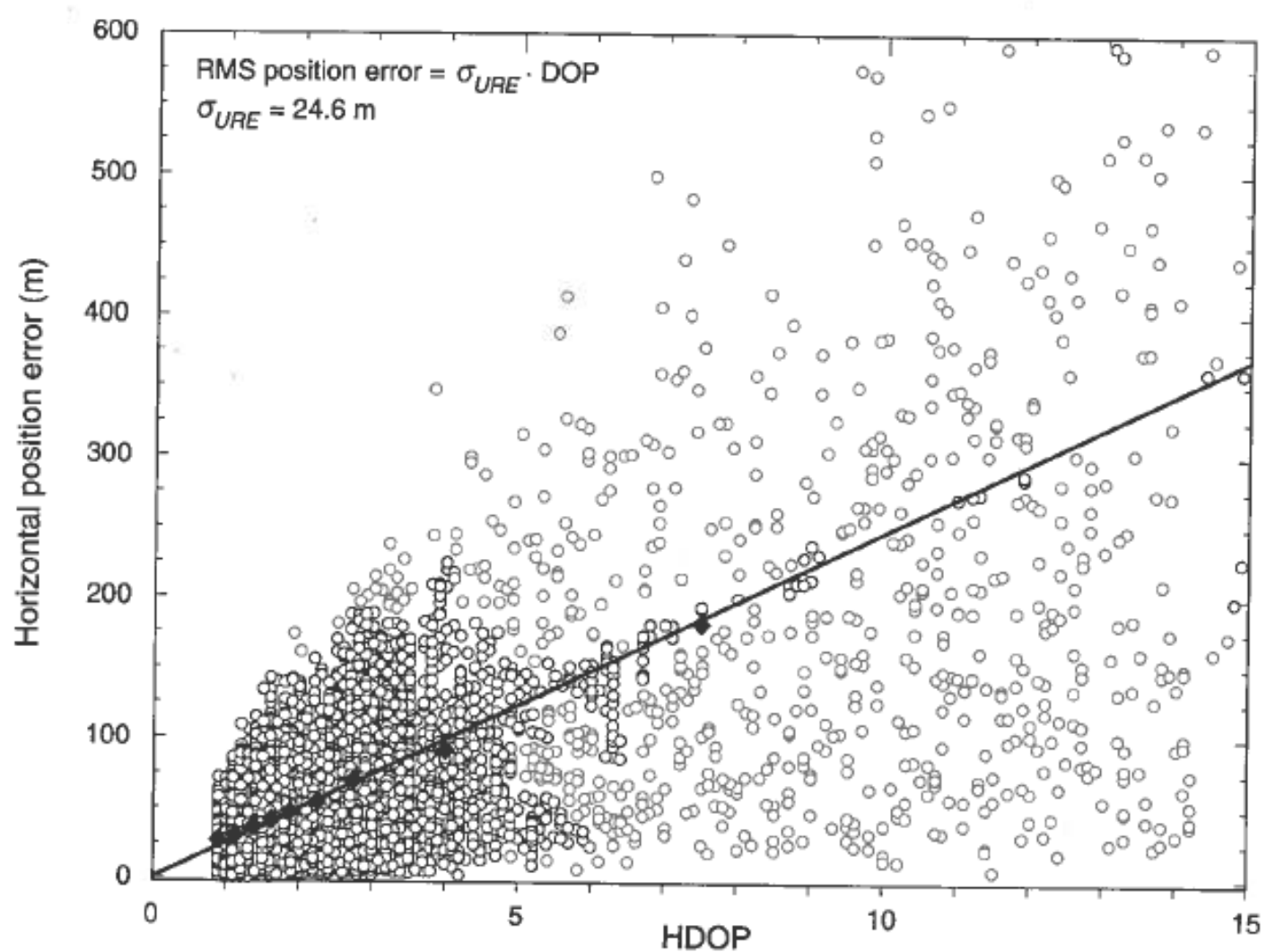
$$GDOP = \sqrt{\text{trace}(H)}$$



Can you express G matrix directly in ENU?

$$G = \begin{bmatrix} -\hat{\mathbf{e}}^{(1)T} & 1 \\ -\hat{\mathbf{e}}^{(2)T} & 1 \\ \vdots & \\ -\hat{\mathbf{e}}^{(K)T} & 1 \end{bmatrix} \quad \hat{\mathbf{e}} = \begin{bmatrix} \cos(el) \sin(az) \\ \cos(el) \cos(az) \\ \sin(el) \end{bmatrix}$$

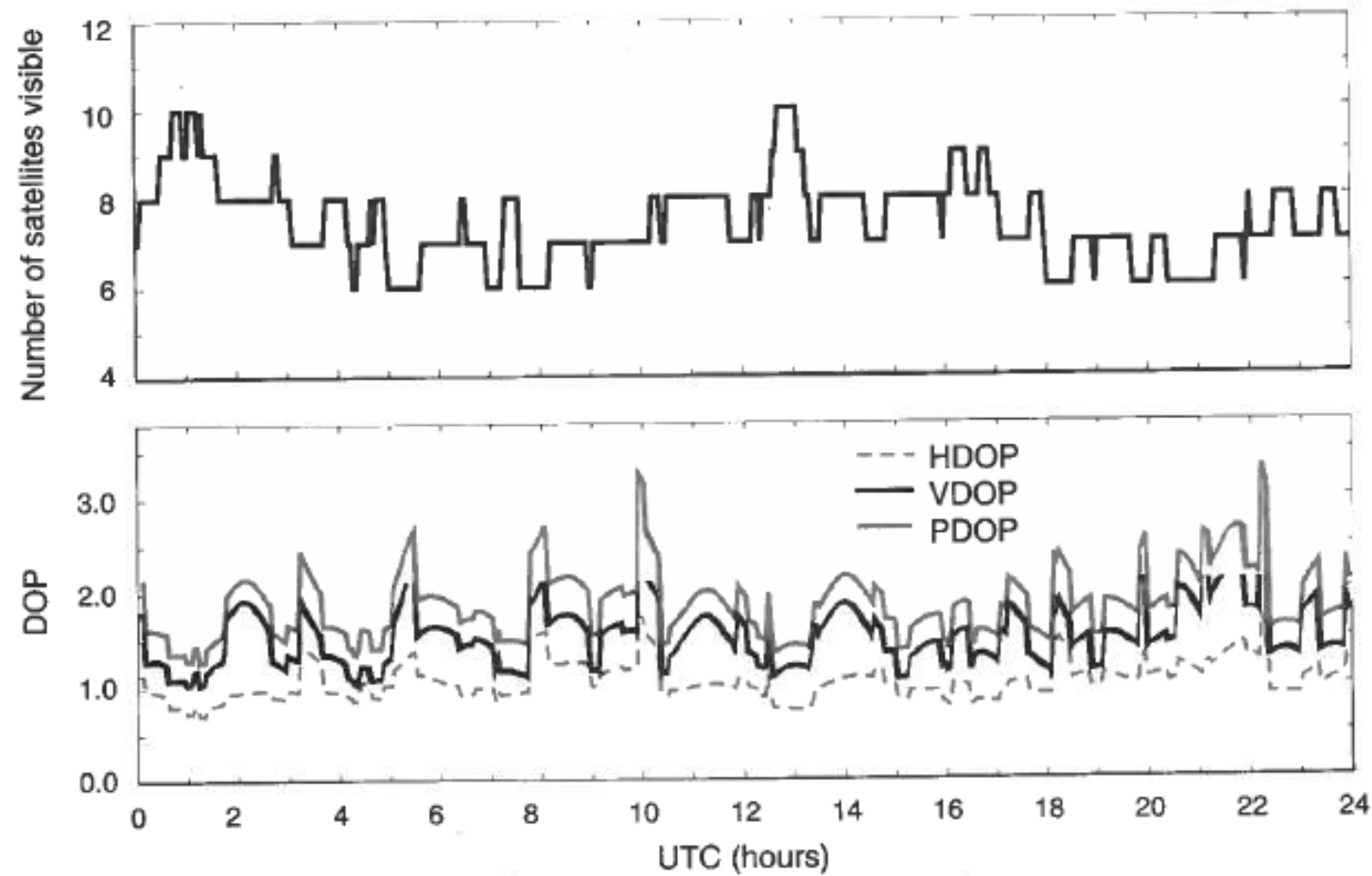
What Does DOP Tell You About Error?



Textbook, P210



Distributions of DOPS



Textbook, P211



Probability Distributions of DOP

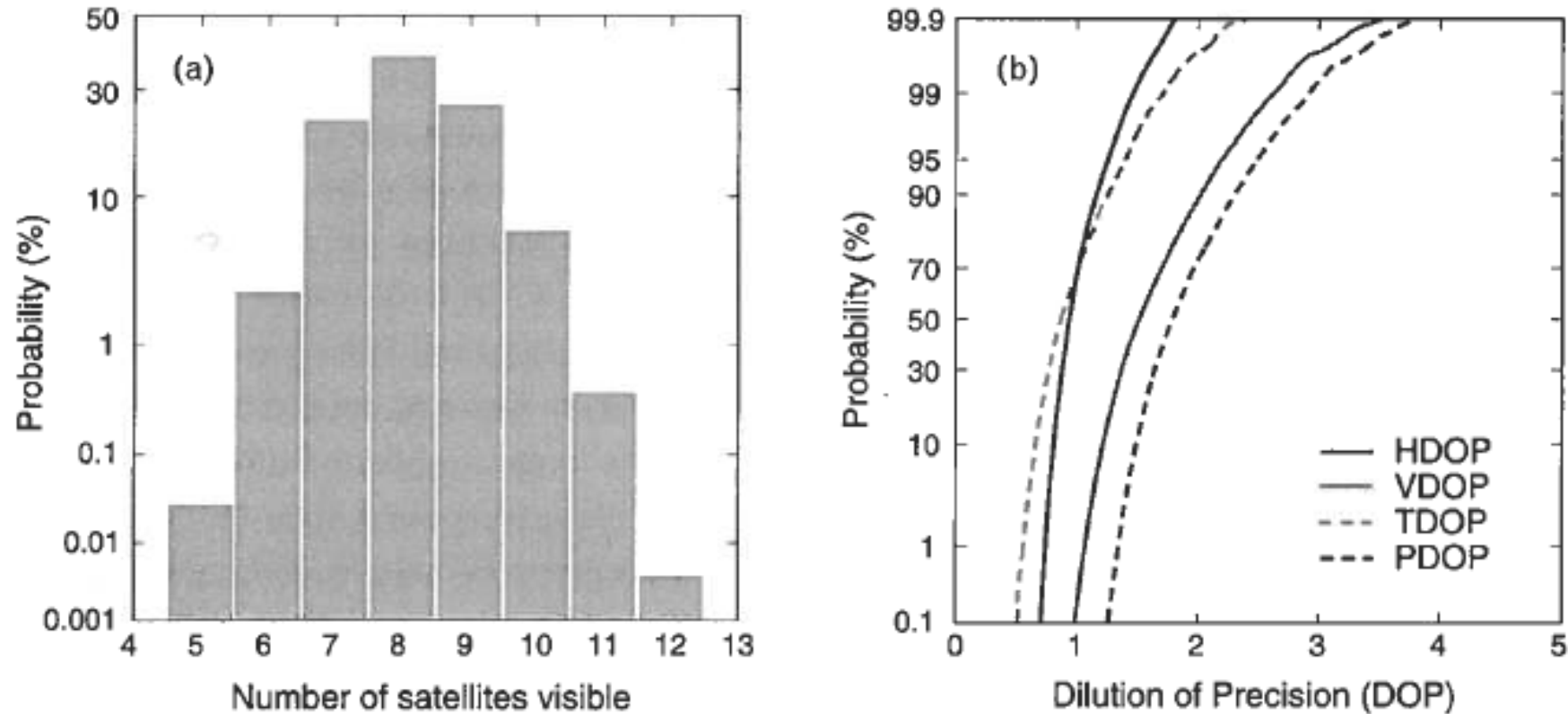


Figure 6.5 (a) Histogram of the number of satellites in view of users worldwide with an unobstructed view of the sky (elevation angle $>5^\circ$) for the 24-satellite baseline constellation. (b) The corresponding cumulative distribution functions of HDOP, VDOP, PDOP, and TDOP values.

Empirical Position Solutions

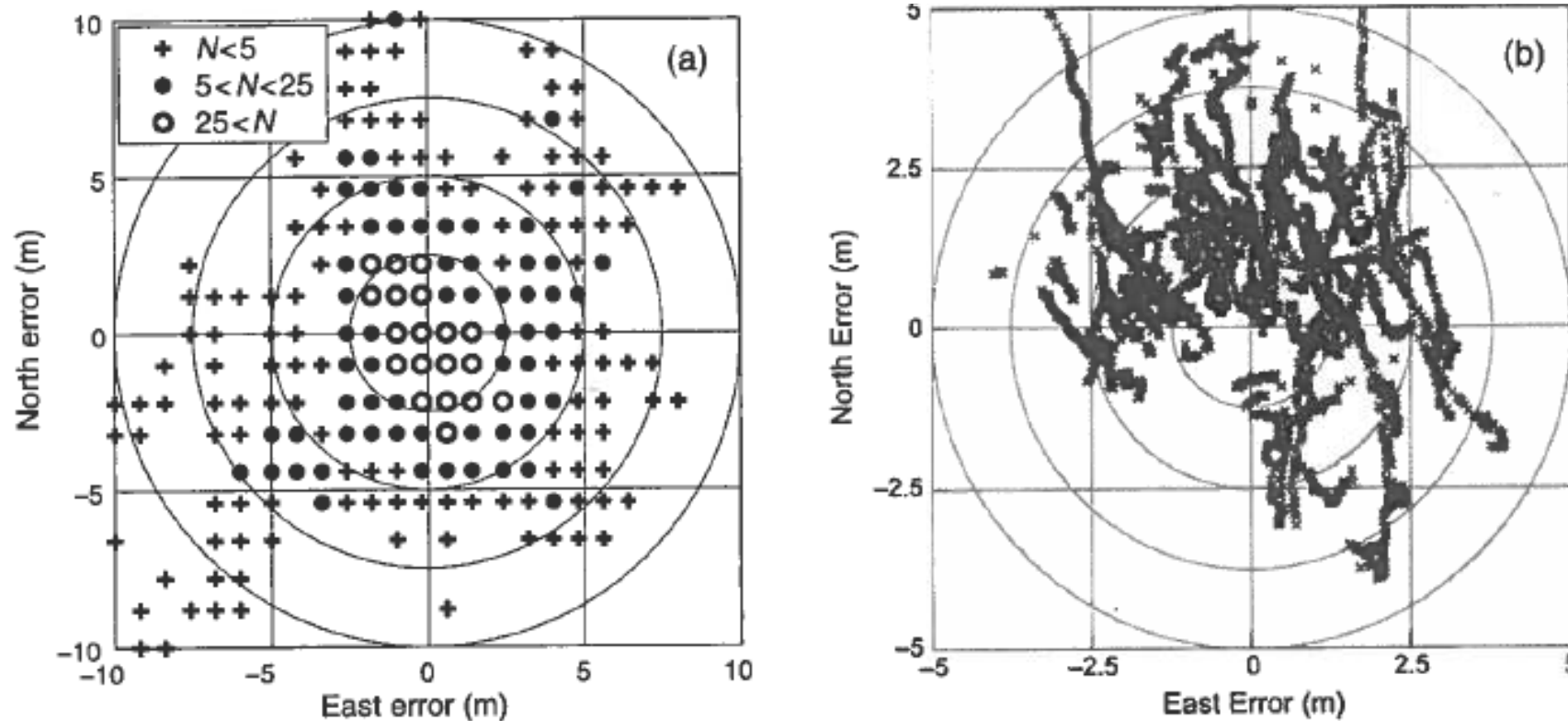


Figure 6.6 Scatter plots of horizontal position error in position estimates obtained from GPS Standard Positioning Service (SPS). (a) Quantized position estimates from an inexpensive receiver for hikers (one-minute samples over 24 hours, N represents the number of outcomes corresponding to each bin). (b) A more capable receiver which gave access to the pseudorange measurements and navigation data (one-second samples over 24 hours), courtesy of Dr. Todd Walter, Stanford University.

DOPPLER MODEL

$$\dot{\rho}(t) = \frac{d(|\vec{r}_{Tx}(t - \Delta t) - \vec{r}_{Rx}(t)| + \delta t_u \cdot c)}{dt}$$

$$\dot{\rho}(t) = \left\{ [\vec{v}_{Tx}(t - \Delta t) - \vec{v}_{Rx}(t)] \cdot \vec{u}_{Tx-Rx} + \frac{d\delta t_u}{dt} \cdot c \right\}$$

In unit of m/s, convert to rad/s by $\cdot 2\pi \cdot f_{L1}/c$

- Doppler modeling is not typically used in navigation but is very useful in remote sensing, e.g., GNSS radio occultation (RO) and reflectometry.
- An accurate Doppler model is essential for increasing coherent integration time, i.e., in order to coherently integrate signal over 1 second, an accurate Doppler model (<0.5 Hz error) is needed.
- Tropospheric delay or bending is considered in a high accuracy or RO modeling case.

IS-GPS-200

Table 20- IV. Broadcast Navigation User Equations (sheet 3 of 4)

Element/Equation	Description
SV Velocity	
$\dot{E}_k = n / (1 - e \cos E_k)$	Eccentric Anomaly Rate
$\dot{v}_k = \dot{E}_k \sqrt{1 - e^2} / (1 - e \cos E_k)$	True Anomaly Rate
$(di_k / dt) = (IDOT) + 2 \dot{v}_k (c_{is} \cos 2\Phi_k - c_{ic} \sin 2\Phi_k)$	Corrected Inclination Angle Rate
$\dot{u}_k = \dot{v}_k + 2\dot{v}_k (c_{us} \cos 2\Phi_k - c_{uc} \sin 2\Phi_k)$	Corrected Argument of Latitude Rate
$\dot{r}_k = eA\dot{E}_k \sin E_k + 2\dot{v}_k (c_{rs} \cos 2\Phi_k - c_{rc} \sin 2\Phi_k)$	Corrected Radius Rate
$\dot{\Omega}_k = \dot{\Omega} - \dot{\Omega}_e$	Longitude of Ascending Node Rate
$\dot{x}'_k = \dot{r}_k \cos u_k - r_k \dot{u}_k \sin u_k$	In- plane x velocity
$\dot{y}'_k = \dot{r}_k \sin u_k + r_k \dot{u}_k \cos u_k$	In- plane y velocity
$\dot{x}_k = -x'_k \dot{\Omega}_k \sin \Omega_k + \dot{x}'_k \cos \Omega_k - y'_k \sin \Omega_k \cos i_k - y'_k (\dot{\Omega}_k \cos \Omega_k \cos i_k - (di_k / dt) \sin \Omega_k \sin i_k)$	Earth- Fixed x velocity (m/s)
$\dot{y}_k = x'_k \dot{\Omega}_k \cos \Omega_k + \dot{x}'_k \sin \Omega_k + \dot{y}'_k \cos \Omega_k \cos i_k - y'_k (\dot{\Omega}_k \sin \Omega_k \cos i_k + (di_k / dt) \cos \Omega_k \sin i_k)$	Earth- Fixed y velocity (m/s)
$\dot{z}_k = \dot{y}'_k \sin i_k + y'_k (di_k / dt) \cos i_k$	Earth- Fixed z velocity (m/s)



Example Velocity Estimation

$$\dot{\rho}^{(k)} = \dot{r}^{(k)} + (\dot{b} - \dot{b}^{(k)}) + \dot{l}^{(k)} + \dot{T}^{(k)}$$

$$\dot{\rho}^{(k)} = (\mathbf{v}^{(k)} - \mathbf{v}) \cdot \mathbf{l}^{(k)} + \dot{b} + \dot{\epsilon}_{\phi}^{(k)}$$

$$\dot{\rho}^{(k)} - \mathbf{v}^{(k)} \cdot \mathbf{l}^{(k)} = -\mathbf{v} \cdot \mathbf{l}^{(k)} + \dot{b} + \dot{\epsilon}_{\phi}^{(k)}$$



$$\dot{\rho}^{(k)} = -\mathbf{l}^{(k)} \cdot \mathbf{v} + \dot{b} + \dot{\epsilon}_{\phi}^{(k)} \quad \Rightarrow \quad \dot{\rho} = \mathbf{G} \begin{bmatrix} \mathbf{v} \\ \dot{b} \end{bmatrix} + \dot{\epsilon}_{\phi} \quad \Rightarrow$$

$$\begin{bmatrix} \hat{\mathbf{v}} \\ \hat{\dot{b}} \end{bmatrix} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \dot{\hat{\rho}}$$

Similar to

$$\delta \rho^{(k)} = -\mathbf{l}^{(k)} \cdot \delta \mathbf{x} + \delta b + \tilde{\epsilon}_{\rho}^{(k)} \quad \Rightarrow \quad \delta \rho = \mathbf{G} \begin{bmatrix} \delta \mathbf{x} \\ \delta b \end{bmatrix} + \tilde{\epsilon}_{\rho} \quad \Rightarrow$$

$$\begin{bmatrix} \delta \hat{\mathbf{x}} \\ \delta \hat{b} \end{bmatrix} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \delta \rho$$

Positioning Techniques

- Standalone PR
 - Current assignment
 - Typical dual frequency accuracy of several m
- Precise PR positions
 - Use precise orbits and clock corrections
- DGPS
 - Corrections for common mode errors between user and reference station sent via communication link
 - Typical accuracy of ~1-3 m

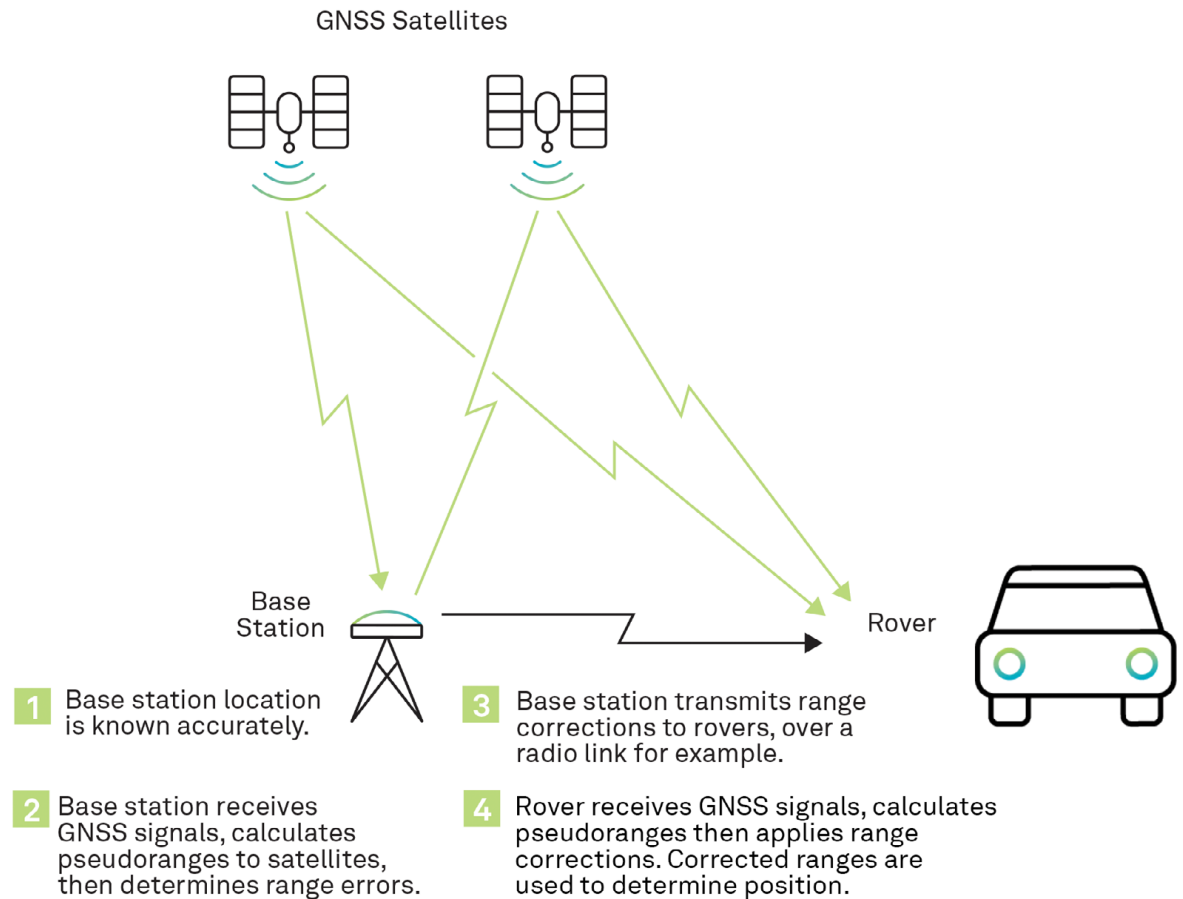
Example Applications

- Standalone PR
 - Many applications such as personal navigation, en route navigation, etc.
- Precise PR positions
 - Mobile phones (via assisted GPS)
- Local and wide area DGPS
 - Transportation
 - Maritime harbor and river navigation
 - Aircraft precision approach and landing
 - Vehicle tracking
- Carrier phase positioning
 - Geodesy, surveying, precision agriculture



Differential GPS (DGPS)

- Pseudorange errors common to receivers in the same area can be reduced using DGPS
- Reference receiver at known location can transmit range corrections (via separate comm link) to nearby users.
- Amount of correction depends on how quickly errors decorrelate in space and time
- In the days of Selective Availability this was a very big deal
- With single frequency receivers this is also important
- Now, clock and orbit errors can be largely corrected using precise data products and dual frequency receivers are becoming more common.



Attribution: © 2015-2023 NovAtel Inc.

Differential GPS

- Pseudorange observables

$$\rho_R = R_R + c(\delta t_R - \delta t^s) + T_R + I_R + M_R + \varepsilon_R$$

$$\rho_U = R_U + c(\delta t_U - \delta t^s) + T_U + I_U + M_U + \varepsilon_U$$

- At reference station, compute pseudorange error $\delta\rho_R = \rho_R - \hat{R}_R = c(\delta t_R - \delta t^s) + O_R + T_R + I_R + M_R + \varepsilon_R$ and send to local users:
- In the user receiver, apply to measurements: $\rho_{Ucorrected} = \rho_U - \delta\rho_R$
$$\rho_{Ucorrected} = R_U + c(\delta t_U - \delta t_R) + (O_U - O_R) + (I_U - I_R) + (T_U - T_R) + (M_U - M_R) + (\varepsilon_U - \varepsilon_R)$$
- Orbit (ephemeris) errors, iono, and tropo experienced by reference and user receivers are similar/correlated.
 - Iono decorrelation is ~1m/100km, tropo depends on height & weather
 - Smaller separation and more frequent corrections yield better accuracy
- Multipath and tracking errors are not correlated between stations

GBAS – Ground Based Augmentation Systems

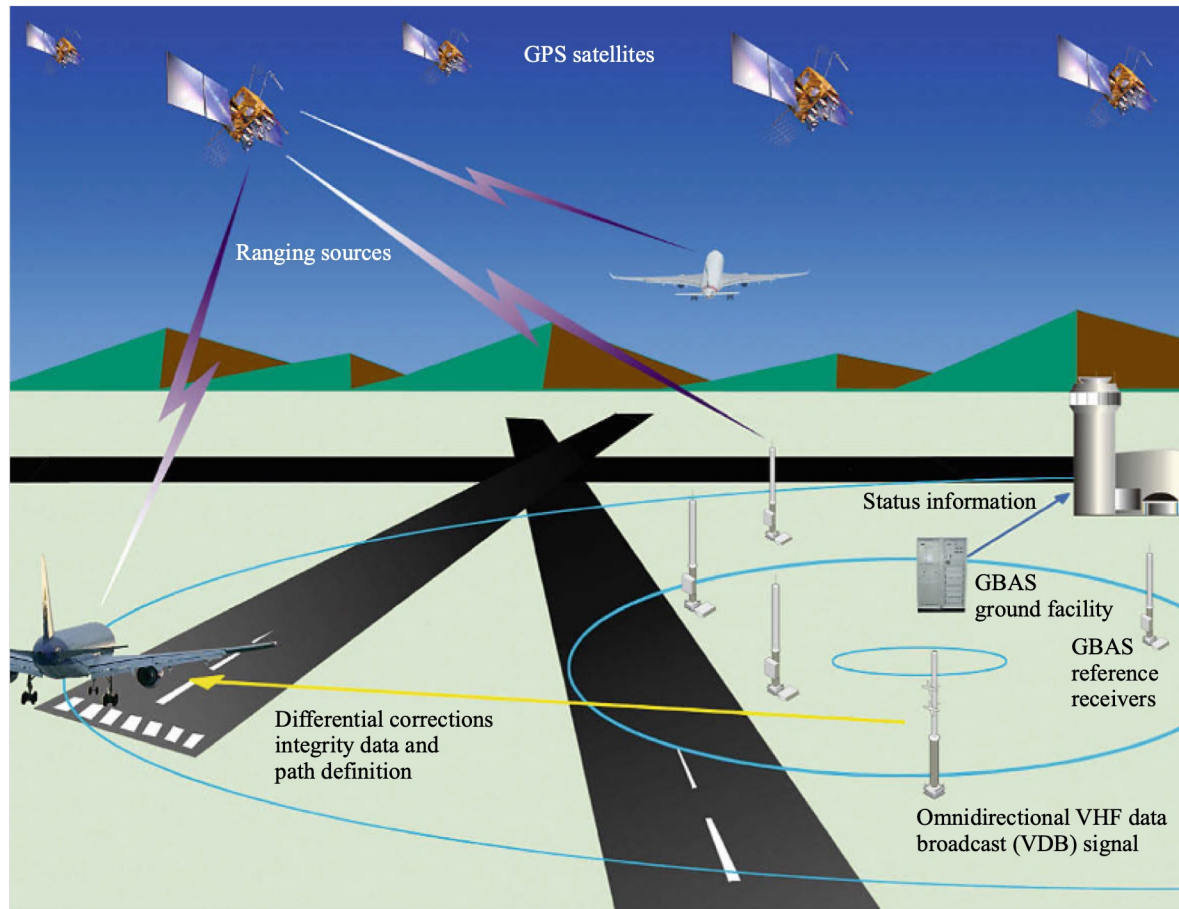
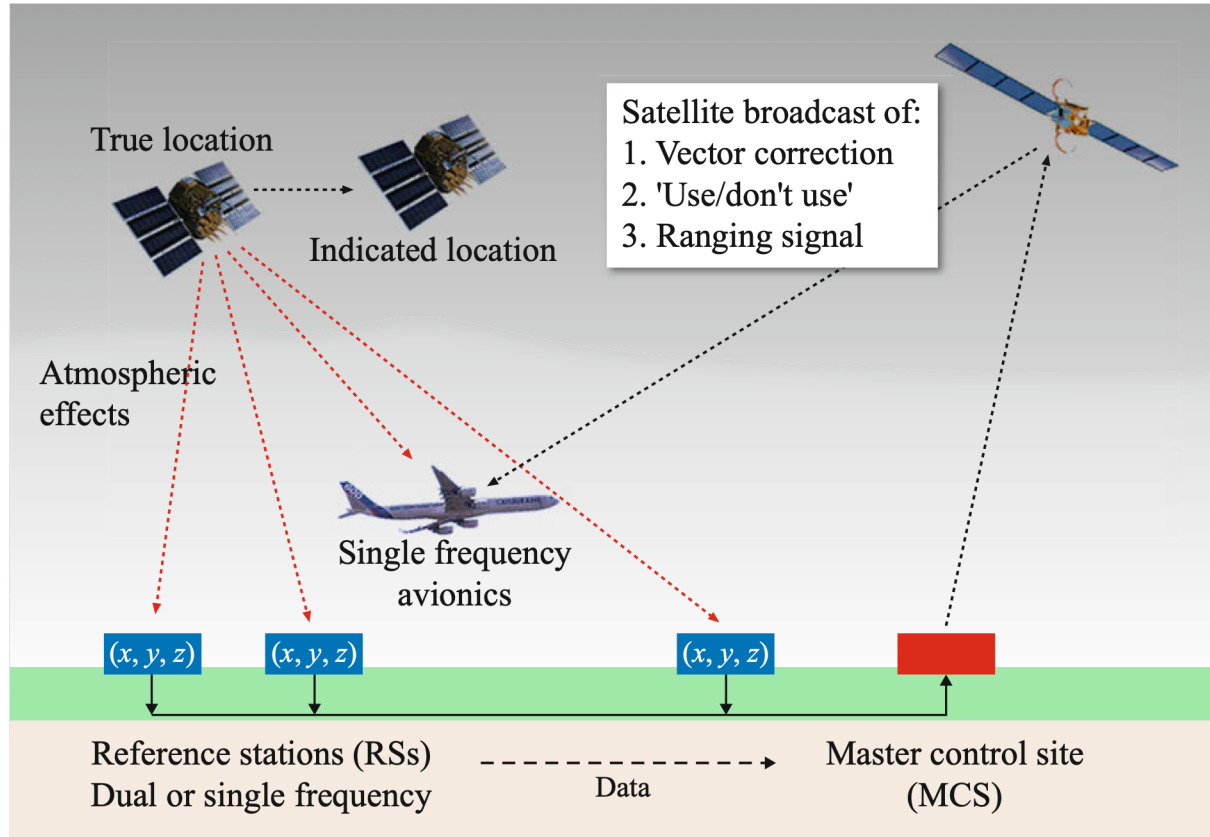


Fig. 31.1 GBAS ground station components-advanced example of terrestrial local area differential GNSS (DGNSS) systems (courtesy of the FAA satellite navigation team)

- Corrections to Pseudorange, carrier phase, reference station
- Anomaly detection
- Transmitted to the aircraft via VHS ILS localizer band 108-118 MHz

S. Pullen, Ch31 in Springer Handbook

SBAS – Space Based Augmentation System



Supports en route and non-precision approach.

LPV – Localizer precision with vertical guidance (LPV)

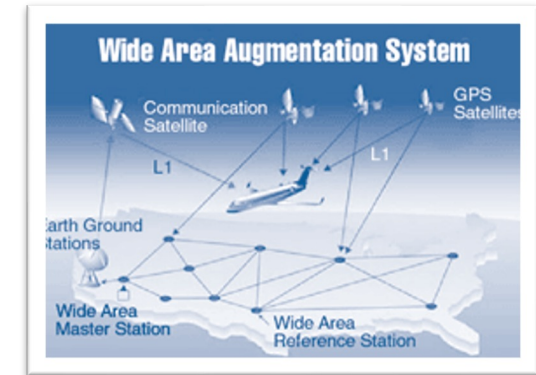
Fig. 30.3 SBAS system architecture. Correction messages are broadcast to the user through a geostationary satellite

Walter (See reference list Springer Handbook)

SBAS – Space Based Augmentation Systems

- Provide precise navigation for aircraft down to 200 ft above the ground over a state's airspace.
- Geographically diverse network of reference stations w/redundant equipment
- Master station(s) compute signal corrections and integrity
- Messages sent to GEO satellites
- GEO satellites transmit on GPS frequency with assigned PRN codes
- **WAAS** has 38 reference stations in CONUS, Alaska, Canada, and Mexico
- Includes both L1 and L5 monitoring
- 3 GEO satellites to transmit signals

European Geostationary Navigation Overlay Service (EGNOS)
Japanese Multi-Functional Satellite Augmentation System (MSAS)

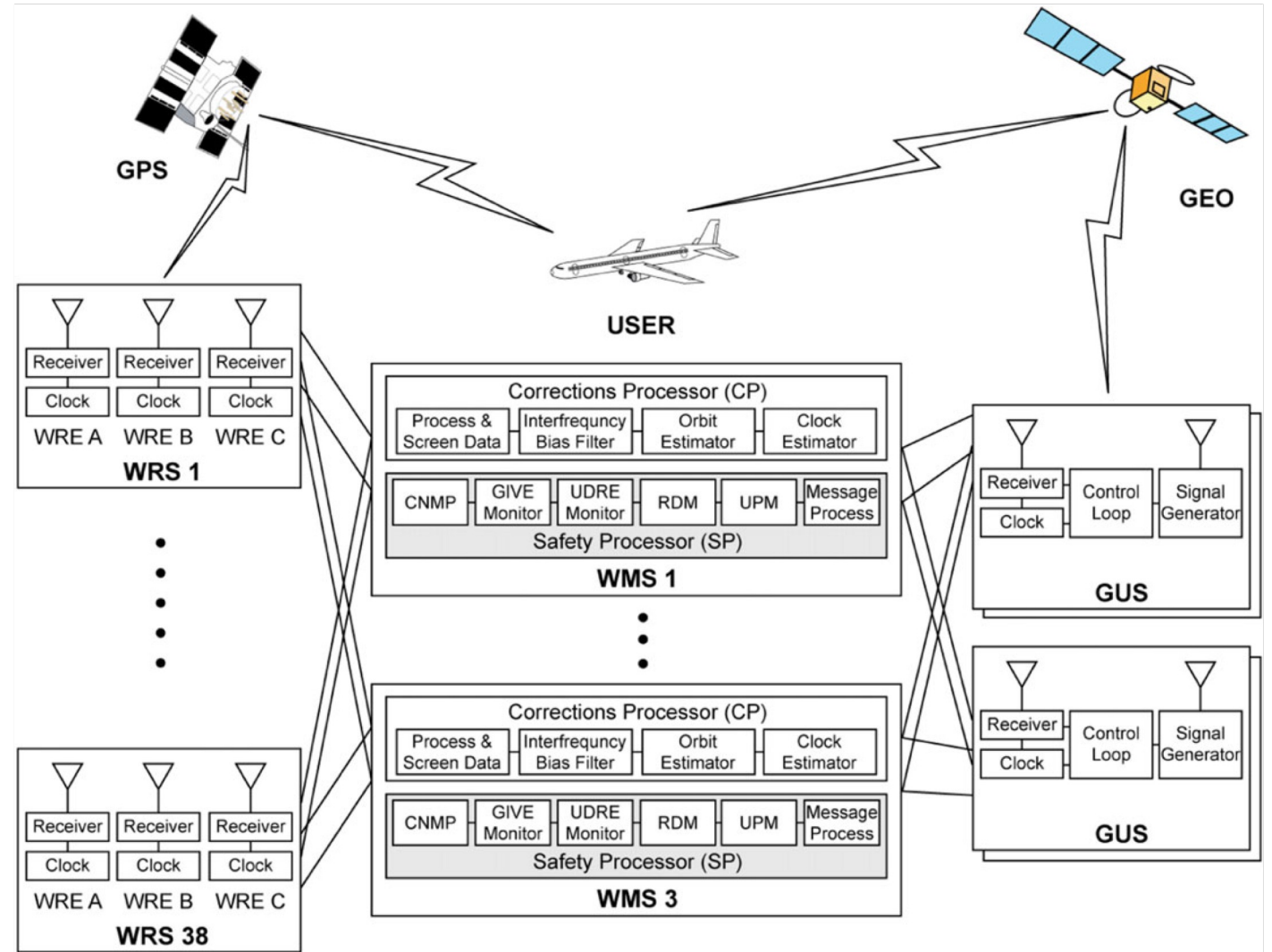


From Walter, et al. *Navigation* 2019, DOI: 10.1002/navi.252

WAAS Architecture

CNMP, Code Noise and MultiPath
GIVE, Grid Ionospheric Vertical Error
GUS, Ground Uplink Station
RDM, Range Domain Monitor
UDRE, User Differential Range Error
UPM, User Position Monitor
WMS, WAAS Master Station
WRE, WAAS Reference Element
WRS, WAAS Reference Station

Walter, et al. WAAS at 15,
Navigation 2019, DOI: 10.1002/navi.252



Integrated GNSS-INS navigation solutions

- Inertial sensors – accelerometers + gyros
- Kalman Filter

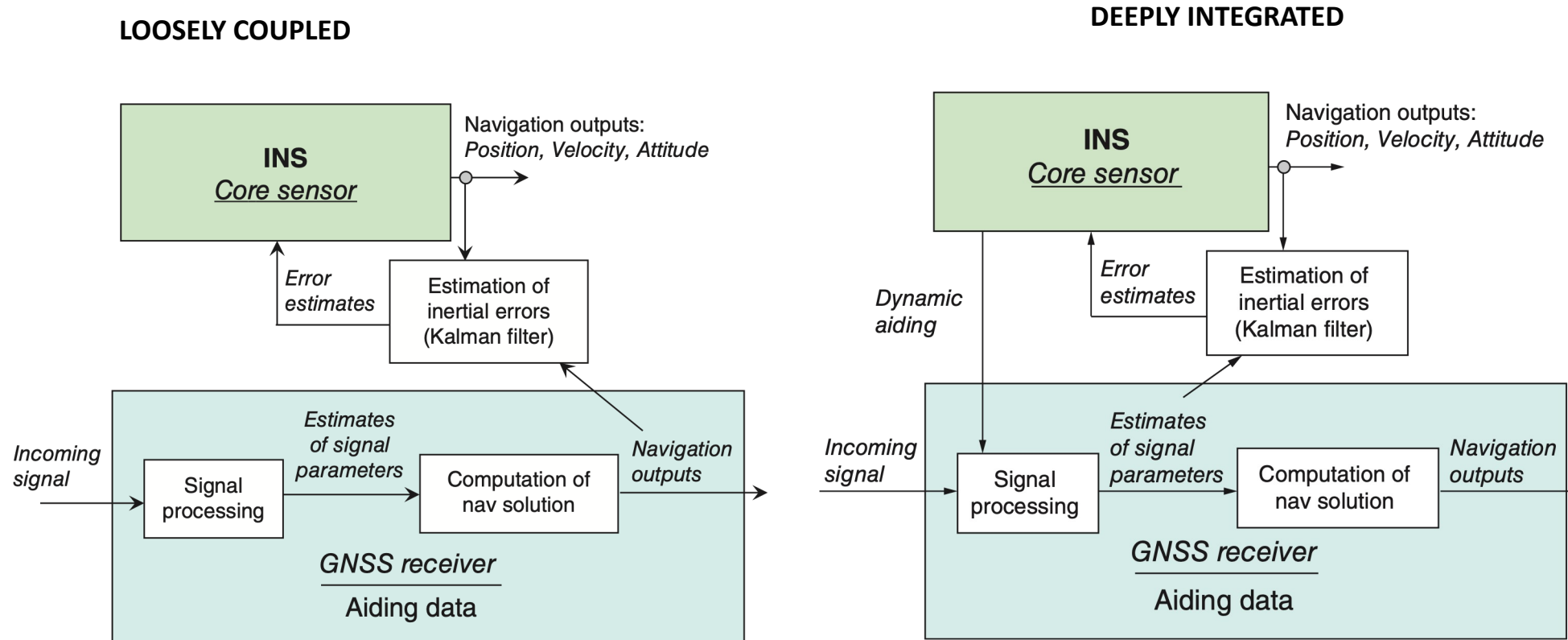


Figure 46.10 Loosely coupled GNSS/INS mechanization. GNSS navigation solution (1) navigation outputs.

Figure 46.18 Deeply integrated GNSS/INS implementation. Deep integration extends tight coupling by incorporating inertial aiding of GNSS signal processing. This enables integration at the signal processing level, thus maximizing the benefits of sensor fusion.

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Next lecture: Precise Positioning

